what is mutex

Implementation

Deadlock situation

Deadlock prevention

↳ Eliminating mutex

↳ " no-preemption

↳ " Hold-wait

↳ " Circular-wait

Semaphore :-

→

Binary Semaphore

Counting Semaphore

user threads

↓

priority high

thread

Binary Semaphore :-

Semcnt : 0 or 1

Semcnt = 1 / 0

job1

wait ( )

// Critical Section

Signal ( )

job2

wait( )

// c.s

Signal( )

if (Semcnt > 0)
{ Semcnt--;
job will cont....
}
else
Blocks this job

Semcnt++;

Semcnt = 0

job1

wait( );

// c.s

Signal( );

job2

Signal( );

// c.s

wait( );

mutex	Binary Semaphore
It is ownership mechanism	It is a signalling mechanism.
mutex object initial value is always 0	Semcnt can be initialized with 0 & 1.
mutex can be used only blw the threads of a process.	Semaphore can be used blw the threads of a process & blw the processes.

→ Counting Semaphore :-

Semcnt = +ve integer

Sem cnt = 3 2 1 0

job1 job2 job3 job4 → wait

wait() wait() wait() wait()

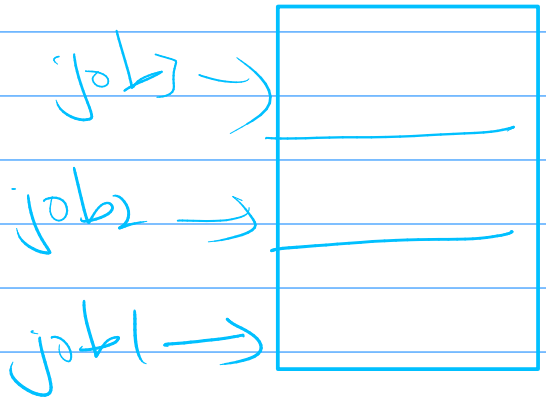
// C.S. // C.S. // C.S. // C.S

Signal() Signal()

KB — Buffer Size

2KB

job1 job2 job3



SpinLock, Record Lock

fcntl

{ mutex
  Semaphore

fcntl (fd, F_DUPFD, 3)

fd is duplicated

new fd. ≥ 3

fcntl (fd, F_DUPFD, 0);

IPC (Inter process comm^n)

PIPE
FIFO
Msg QUEUE
Shared Memory

} process - process comm^n